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#include <Wire.h> // For I2C communication
#include <LiquidCrystal_I2C.h> // For the LCD display
#include <Adafruit_MPU6050.h> // For the MPU6050 sensor
#include <Adafruit_Sensor.h> // Required helper for Adafruit sensors

// Initialize the LCD with address 0x27, 16 columns, 2 rows
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Create an MPU6050 object
Adafruit_MPU6050 mpu;

// Vibration Sensitivity Thresholds
float lightVibration = 0.2; // Anything above 0.2g is light
float heavyVibration = 1.0; // Anything above 1.0g is heavy

void setup() {
  Serial.begin(115200); // Start serial monitor for debugging

  // 1. Setup LCD
  lcd.init(); // Turn on the LCD
  lcd.backlight(); // Turn on the backlight
  lcd.setCursor(0, 0); // Go to column 0, row 0
  lcd.print("System Starting"); // Print welcome message

  // 2. Setup MPU6050
  if (!mpu.begin()) {
    lcd.setCursor(0, 1);
    lcd.print("Sensor Error!");
    while (1);
  }

  mpu.setAccelerometerRange(MPU6050_RANGE_4_G);
  delay(2000);
  lcd.clear();
}

void loop() {
  sensors_event_t a, g, temp;
  mpu.getEvent(&a, &g, &temp);

  float x = a.acceleration.x / 9.8;
  float y = a.acceleration.y / 9.8;
  float z = a.acceleration.z / 9.8;

  float totalAccel = sqrt((x * x) + (y * y) + (z * z));
  float vibration = abs(totalAccel - 1.0);

  lcd.setCursor(0, 0);
  lcd.print("Vib: ");
  lcd.print(vibration, 2);
  lcd.print(" g");

  lcd.setCursor(0, 1);

  if (vibration < lightVibration) {
    lcd.print("Status: Calm");
  }
  else if (vibration >= lightVibration && vibration < heavyVibration) {
    lcd.print("Status: Mod Vib");
  }
  else {
    lcd.print("Status: DANGER!");
  }

  delay(100);
}

```